VW–Fiat: demand, substitution and merger effects

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1. Research question and principal findings

Would a merger of VW and Fiat harm car buyers? I compare an uninstrumented simple logit with a random-coefficients BLP model, using identical demand controls and fixed effects. A jointly estimated homogeneous logit provides an additional comparison. Each historical country-year is a separate hypothetical merger; no actual historical merger is asserted.

OLS predicts a 0.930% average increase in merging firms' prices, versus 0.661% for BLP. However, OLS implies negative marginal costs for 4,544 of 6,000 products. Its counterfactual is therefore a diagnostic, not a credible policy forecast. BLP yields positive costs throughout and predicts stronger VW–Fiat diversion. Its precision relies materially on the additional supply restrictions.

2. Data and modelling choices

The course data contain the 40 highest-selling models in each of 150 markets: Belgium, France, Germany, Italy and the UK, 1970–1999. All 6,000 rows are retained. Ownership follows the historical firm codes: VW is firm 26 and Fiat firm 4, including their assigned brands. The outside option is buying none of the included cars, encompassing used cars, omitted new cars and no purchase. It is not a price-setting firm.

$M_t = pop_t/2.5$; $s_{jt} = q_{jt}/M_t$; $s_{0t} = 1 - \Sigma_j s_{jt}$.
The household-equivalent market assumes 2.5 people per potential annual purchase opportunity; it is not a measured household series. Inside shares average 7.24% (range 3.85–10.00%). Population as market size is the main robustness check. Unit demand omits fleet and multiple-household purchases.

Price p is gross local-currency price divided by nominal GDP per capita Y, the supplied princ variable. This avoids pooling currency levels and permits a coherent money-metric welfare calculation with linear price utility. GDP per capita is a normalisation, not observed household income. The counterfactual holds Y fixed. Top-40 selection omits individually adjusting competitors and can also threaten instrument validity. [1]

3. Demand and the econometric distinction

$u_{ijt} = \lambda_t + \eta_{b(j)} + x'_{jt}\beta - \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt}$.

The outside systematic utility is zero; ε is independent type-I extreme value with scale normalised to one. Controls x are weight (tonnes), fuel consumption (litres/100 km), domestic origin, power-to-weight (100 kW/tonne), and footprint (m²). Higher fuel consumption means worse fuel economy. Brand effects absorb persistent brand quality; country-year effects absorb common inside-good demand. Price identification uses differences remaining after these effects and controls.

In simple logit α is common. Choice probabilities equal $\exp(\delta_{jt})/[1 + \Sigma \exp(\delta_{kt})]$. Their inverse gives a linear estimating equation:

$\delta_{jt} = \log(s_{jt}/s_{0t}) = \lambda_t + \eta_{b(j)} + x'_{jt}\beta - \alpha p_{jt} + \xi_{jt}$.

The requested baseline estimates this equation by OLS, with no excluded instruments or supply moments. Consistency requires regressors, including residualised price, to be orthogonal to ξ. Unobserved quality can raise both demand and equilibrium price, making the OLS price coefficient less negative. Fixed effects do not remove model-specific, time-varying quality. The estimated slope is −1.108 (SE 0.179), versus −2.775 (0.611) in demand-only 2SLS with the same controls. This difference is consistent with endogeneity, not proof of instrument validity. Demand-only IV still implies 132 negative costs in 72 markets.

The merger simulation retains estimated residual quality ξ, so baseline probabilities reproduce observed shares. Using only fitted controls and fixed effects would discard that quality and change the starting market.

BLP: $\alpha_i = \exp(\mu + \sigma\nu_i)$, $\nu_i \sim N(0,1)$; $s_{jt} = \int P_{ijt}\, dF(\nu)$.

Only price sensitivity is random, keeping the extension small and demand downward sloping for every consumer. Aggregating individual logits relaxes aggregate independence of irrelevant alternatives (IIA): products attract different mixtures of price-sensitive buyers. It does not introduce random preferences for size or domestic origin. Segment nests or additional random coefficients would impose different substitution patterns. The common distribution in affordability units is a maintained restriction. [1,2]

4. Identification, estimation and inference

BLP inverts observed shares for mean utilities at each candidate (μ, σ), recovers demand shocks ξ, and uses multiproduct Bertrand conditions to recover costs. Log effective marginal costs depend on the same five controls, brand and market effects, and log(avcpr/avexr), an exporter-price/exchange-rate cost proxy. Cost residuals are ω. This is an additional economic restriction, not just an estimation convenience.

$g(\theta) = [Z'_D\xi/N, Z'_S\omega/N]'$; $\hat{\theta} = \arg\min_\theta N\, g(\theta)'Wg(\theta)$.

For each of power, size and fuel, the excluded instruments sum squared characteristic distances to other products of the same firm and to rivals: six differentiation instruments. They shift competitive conditions. Exclusion requires orthogonality to demand shocks and, for supply moments, to cost shocks. Endogenous positioning, common platforms and selected assortments threaten these assumptions. All fixed effects are absorbed consistently. VAT is market-constant and hence absorbed; it is not an excluded instrument. [2,3]

There are 11 demand and 12 supply moments, and 13 slopes/distribution parameters after absorption. Two-step GMM uses a car-model-clustered weighting matrix; standard errors also cluster by the 285 model codes. The linear first-stage partial R^2 is 0.128. Its robust Wald $\chi^2(6)$ is 91.68, giving an F-style Wald/6 of 15.28. The raw 91.68 is not an F-statistic and does not establish nonlinear identification or exclusion validity.

BLP estimates μ = 2.516 (0.439), σ = 0.527 (0.118), and mean α = 14.216. Hansen J = 11.58 on 10 degrees of freedom (p = 0.314); nonrejection does not validate the assumptions. With demand alone, σ = 0.241 (1.543). Thus supply assumptions are central to precision. The joint homogeneous logit estimates α = 5.026 (0.511), with J p = 0.029. Its moments fit less comfortably.

Forty deterministic quadrature nodes, doubled to eighty, and three starting values give stable BLP results. Share matching follows from inversion, not predictive success. Boundary σ = 0 and exploratory model selection preclude naive significance claims. Cost-proxy γ = 0.0023 (0.0025) is imprecise; full cost estimates are supplied in the CSV outputs. [4]

5. From demand to merger prices and welfare

Let $D_{jk} = \partial s_j / \partial p_k$ and H_{jk} indicate common ownership. A firm maximises the sum of product profits. Its price first-order conditions and their inversion are:

$$s + (H \odot D)'(p - \tilde{c}) = 0; \quad \tilde{c} = p + [(H \odot D)']^{-1} s.$$

Market size cancels because it multiplies every term. With market-constant VAT τ, effective cost $\tilde{c} = (1 + \tau)c/Y$ yields these same conditions; c is producer cost in local currency. Ownership changes merge firm 4 into 26. Holding product quality and costs fixed, all firms reoptimise prices, including rivals. This is static Bertrand competition with constant marginal costs, no entry, capacity constraints or coordination.

$\varepsilon_{jk} = D_{jk} p_k / s_j$; diversion $j \to k = -D_{kj}/D_{jj}$.
Under simple logit, own elasticity is $-\alpha p_j (1 - s_j)$, cross elasticity is $\alpha p_k s_k$, and diversion is $s_k/(1 - s_j)$. With firm share S_f, every product has absolute markup $1/[\alpha(1 - S_f)]$. OLS therefore implies median markup/price of 133.5% and negative costs for 75.7% of observations, in all 150 markets. Costs are neither clipped nor replaced. The computed OLS counterfactual solves the equations but fails economic admissibility.

$$CS_t = \int \log[1 + \Sigma_j \exp(V_{ijt})]/\alpha_i \, dF(i).$$

This money-metric surplus is per potential household in GDP-per-capita units, under quasi-linearity. Multiply ΔCS by Y for local currency and by M for the market total. For BLP, integrate each individual log-sum divided by their own α; dividing by mean α is incorrect. Do not sum different nominal currencies or years. [4]

For admissible models, bisection finds the common proportional reduction in all merging-firm costs that restores baseline consumer surplus, with rivals' costs unchanged. This is a consumer-surplus threshold, not total welfare. OLS has no economically meaningful percentage threshold here: reducing a negative cost proportionally would increase it.

Checks of derivatives, price recovery, merger first-order conditions, unilateral deviations, integration and starting values passed. These are local checks, not proof of global equilibrium. No merger confidence intervals are supplied. BLP validation completed in stages; OLS completed in one run.

6. Comparing results and testing robustness

Tables 1–2 compare OLS, joint homogeneous logit and BLP. BLP's median own-price elasticity is −3.99, versus −0.76 for OLS. OLS estimates a negative power-to-weight slope; BLP's is positive. Fuel effects are imprecise. Domestic origin is strongly associated with demand in both; BLP's coefficient 1.243 multiplies individual odds versus the outside by exp(1.243) ≈ 3.47, not market shares by that amount.

OLS and joint logit have identical VW→Fiat diversion of 1.10%, because baseline shares are identical and homogeneous diversion excludes α. BLP gives 4.39% (reverse: 3.01%). The 1999 median off-diagonal cross elasticity is 0.0010 for OLS, 0.0046 for joint logit and 0.0202 for BLP. This documents genuinely different substitution patterns, although BLP still restricts how similarity matters.

Merger price increases are 0.930% under OLS, 0.372% under demand-only IV, 0.205% under joint logit and 0.661% under BLP. Under homogeneous logit, calibrated to the same baseline shares and prices, α times each absolute merger price change is invariant to α. Hence 0.930 × 1.108 ≈ 0.205 × 5.026. This identity is verified numerically. The OLS/joint-logit difference comes from price sensitivity, whereas BLP also changes diversion and pass-through; it is not a pure causal decomposition of heterogeneity.

Consumer-surplus percentages require care. OLS loses 0.212% and BLP 0.642% of their respective model-implied baseline surpluses. Yet their money-metric losses are 1.552 and 1.288 basis points of GDP per capita per potential household, respectively. Percentage losses alone cannot rank money losses across models. OLS and joint logit have identical percentage losses because their homogeneous utilities change identically in α-scaled prices.

Italy has the largest BLP mean price increase (1.413%). Mean BLP break-even savings are 0.805%, ranging across markets from 0.105% to 3.626%; 5% savings raise aggregate consumer surplus in every baseline market. These are hypothetical efficiencies, not evidence they are achievable. Population market size lowers BLP's price effect to 0.404%, versus 0.369% for OLS. Log-price OLS has no finite unconstrained Bertrand equilibrium; log-price IV has weak instruments (F-style 0.98). Tables 3–5 report these robustness checks.

7. Theoretical questions 2.1–2.3

2.1 Homogeneous products. Full-information Bertrand cannot explain simultaneous sales of identical goods at unequal prices: buyers choose the cheapest supplier. Equal-price ties can allocate unequal shares but do not explain price dispersion. Cost differences alone cannot give expensive sellers positive sales with unlimited capacity. Annual averages obscure contemporaneous prices, preventing direct rejection from annual price dispersion alone.

Alternative course models permit dispersion and positive shares. Varian gives firms captive uninformed customers; asymmetric captive masses could capture domestic distribution advantages. Burdett–Judd and Stahl allow costly search; Butters limits information through advertising. Levitan–Shubik permits expensive firms to serve demand left unfilled by capacity-constrained rivals. Extensions to many firms with asymmetric reach or capacity can explain unequal shares. Symmetric versions do not automatically explain persistent home bias. Diamond's identical-firm monopoly-price equilibrium alone does not explain dispersion. [1]

2.2 Perloff–Salop differentiation. Consumers draw independent product match values; utility is match value minus price. Horizontal differentiation comes from individual rankings, with otherwise symmetric firms. The lecture assumes compulsory purchase. Both empirical models retain match tastes but add an outside option and characteristic, brand and unobserved-quality shifts. BLP also makes price sensitivity heterogeneous. Shared characteristics do not imply correlated tastes in OLS.

2.3 A testable restriction. Adapt Perloff–Salop to Gumbel tastes, an outside option and market intercepts. Estimate the logit inverse by OLS and jointly test that five characteristic slopes and 35 non-reference brand effects are zero, leaving market effects and the price slope unrestricted:

H_0: $\beta_{weight,fuel,home,power,size} = 0$ and $\eta_b = 0$ for all non-reference b.

The model-clustered Wald statistic is $\chi^2(40) = 2{,}204.92$, $p < 0.001$. This rejects the specified no-systematic-differentiation restriction under the regression's orthogonality assumptions. It is not a distribution-free rejection of every Perloff–Salop variant: endogenous prices, misspecified tastes or correlated aggregate quality errors can also invalidate the maintained regression. Code checks all 40 restrictions.

8. Theoretical question 2.4 and policy assessment

2.4 Many firms. In the symmetric single-product Perloff–Salop model, entry's effect depends on the tail of the match-value distribution. With n firms, Gumbel scale τ and utility $-\alpha p$, logit gives markup $\tau/[\alpha(1 - 1/n)]$, converging to $\tau/\alpha > 0$. Competition need not eliminate markups. For normal tastes with standard deviation τ, extreme-value spacing instead gives markup asymptotically proportional to $\tau/[\alpha\sqrt{(2 \log n)}]$, which vanishes. Thin normal tails make the best alternatives increasingly close; exponential Gumbel tails preserve a nonvanishing gap. [1,5]

Thus taste-distribution assumptions are economic restrictions, not harmless computational choices. Price coefficients are identified relative to normalised taste scale; changing that distribution can change inferred substitution and market power. Random price sensitivity relaxes aggregate IIA but does not remove the conditional Gumbel assumption. The many-firm thought experiment also requires firms' portfolio shares to vanish; adding products to a few large owners is different.

Policy assessment. I would not approve the merger unconditionally: BLP predicts price increases without efficiencies, particularly in Italy. OLS demonstrates why a negative price slope alone is insufficient for merger advice. Approval would require credible merger-specific savings, evidence on omitted competitors and a defensible conduct model. The small mean efficiency threshold does not guarantee every market or consumer benefits.

References and AI disclosure

[1] AIO 2026: Project 1; Lecture 1, slides 31–36; Lectures 5–6; demand.pdf, logit.pdf and 4-demand-handout.pdf.

[2] Berry, Levinsohn & Pakes (1995), Econometrica 63, 841–890. [3] Gandhi & Houde (2019, revised 2020), NBER WP 26375.

[4] PyBLP 1.3.0 documentation: pyblp.readthedocs.io (Problem, differentiation instruments, prices and consumer surplus). [5] Perloff & Salop (1985), Review of Economic Studies 52, 107–120.

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Output 1. Estimates and aggregate counterfactuals

Table 1. Demand estimates (model-clustered SEs). All models include country-year and brand fixed effects.

Parameter	OLS logit	Joint logit	BLP
Price slope	-1.108 (0.179)	-5.026 (0.511)	—
Weight	-0.025 (0.313)	3.024 (0.575)	3.041 (0.906)
Fuel use	-0.026 (0.022)	-0.058 (0.027)	-0.041 (0.027)
Domestic	1.222 (0.049)	1.213 (0.053)	1.243 (0.050)
Power/weight	-0.619 (0.277)	1.927 (0.445)	2.028 (0.771)
Footprint	0.090 (0.062)	0.201 (0.068)	0.312 (0.092)
μ (log α)	—	—	2.516 (0.439)
σ (log α)	—	—	0.527 (0.118)

OLS has no instruments. Joint logit and BLP use IV and supply moments. BLP's price slope is $-\exp(\mu + \sigma\nu)$.

Table 2. Baseline household market; 150 country-years

Statistic	OLS logit	Joint logit	BLP
Median own ε	-0.755	-3.424	-3.993
Negative costs	4,544	0	0
Merging price, %	0.930	0.205	0.661
VW → Fiat, %	1.101	1.101	4.386
CS change, %	-0.212	-0.212	-0.642
ΔCS, GDPpc bp	-1.552	-0.342	-1.288
Cost saving, %	N/A	0.287	0.805

Prices/diversion are sales-weighted within markets, then averaged equally across markets. GDPpc bp = 0.01% of GDP per capita per household. OLS's invalid costs preclude a credible efficiency threshold.

Output 2. Country differences and robustness

Table 3. Equal-weight averages over 30 years per country. Prices refer to merging firms. Savings restore aggregate consumer surplus.

Country	OLS price %	BLP price %	BLP saving %
Belgium	0.869	0.559	0.639
France	0.764	0.429	0.512
Germany	1.205	0.637	0.720
Italy	1.520	1.413	1.863
UK	0.292	0.266	0.293

Table 4. Alternative specifications

Specification	Price rise, %	CS change, %
OLS: population	0.369	-0.086
BLP: population	0.404	-0.454
BLP: local IVs	0.808	-0.711
BLP: brand-country FE	0.724	-0.726

All alternatives re-estimate demand. OLS population still has 4,517 negative costs. Local IVs use characteristic proximity counts. Brand-country effects absorb domestic origin, so that regressor is omitted in that robustness.

Table 5. Matched log-price specifications

Log price	Slope (SE)	Median own ε	Negative costs
OLS	-0.859 (0.219)	-0.858	6,000
2SLS	-6.918 (4.402)	-6.910	0

Same controls, fixed effects and model clustering. Log OLS has $\alpha < 1$, implying unbounded profits as price tends to infinity. Log IV gives a 0.106% merger price rise, but its first-stage partial R^2 is only 0.0041 and F-style statistic 0.98; treat this as a weak-IV diagnostic. No monetary surplus is obtained by dividing a log-sum by the log-price slope.